Predictive Analytics for Egypt's Food Market

Level 3 (2025-2026)

Data Science (CS437)

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Problem Statement:

In recent years, the Egyptian food market has experienced unprecedented price volatility, driven by a complex interplay of internal market dynamics and external macroeconomic shocks. Conventional forecasting methods often fail to account for the non-linear impact of rapid currency devaluation, fuel price adjustments, and global geopolitical crises. This high degree of uncertainty poses a significant challenge for supply chain stability, consumer purchasing power, and national food security.

The core problem lies in the difficulty of capturing how these diverse factors ranging from seasonal religious occasions (such as Ramadan) to global events (like the COVID-19 pandemic and the Russia-Ukraine war) interact to influence food commodity prices.

This project aims to bridge this gap by developing a sophisticated data-driven predictive framework. By integrating historical price trends with key economic indicators including USD exchange rates and energy costs the system utilizes machine learning to model these intricate relationships. The ultimate goal is to provide accurate, multi-dimensional price forecasts that empower decision-makers and stakeholders to navigate economic instability with data-backed insights.

Dataset Description:

1. Primary Source and National Representativeness

The data was officially sourced from the Central Agency for Public Mobilization and Statistics (CAPMAS) in Egypt (https://www.capmas.gov.eg). As the nation's primary statistical authority, CAPMAS provides data that serves as a representative sample of the entire Egyptian geography. Utilizing this source ensures that the model is trained on authoritative, government-verified records that reflect actual market conditions across various governorates.

2. Temporal Scope and Longitudinal View

Timeframe: The dataset covers an extensive period of nearly eight years, starting from June 2018 and extending through January 2026.

Granularity: Recorded at monthly intervals, this granularity allows the model to capture cyclical seasonality (such as Ramadan and harvest seasons) and the immediate effects of sudden economic policy changes.

3. Dataset Volume and Composition

Scale: The raw dataset consists of 5,652 observations, providing a solid statistical base for training predictive models.

Diversity: It covers a wide range of essential commodities including Staple Goods (Rice, Sugar, Flour), Animal Protein (Meat, Poultry, Seafood), and Fresh Produce (Fruits and Vegetables).

4. Real-World Complexity

A key value of this dataset is its Real-World Complexity. Since it was integrated from raw official reports, it contains the natural noise and structural challenges (such as inconsistent labeling and varied encodings) typical of genuine economic data. This complexity ensures that the project moves beyond theoretical exercises into practical, data-driven problem-solving.

Methodology & Data Preprocessing Pipeline

Stage 1: Data Acquisition & Integration

A. Data Extraction (The Origin):
The first step in our pipeline was the Extraction of raw historical data from the official portal of the Central Agency for Public Mobilization and Statistics (CAPMAS). We exported the food price records from multiple official reports into structured CSV files. This process allowed us to move from unstructured government publications to a machine-readable format.

B. Data Integration & Consolidation:
After extracting the files, we performed Data Integration by combining two primary sources: 2018-2026.csv (containing long-term records) and egypt_prices_all.csv (containing detailed recent prices). Using Python's Pandas library, we standardized the column headers and utilized the pd.concat() function to merge these datasets into a single, unified DataFrame.

Stage 2: Data Cleaning & Structural Correction

Raw data is often noisy and inconsistent; therefore, a rigorous cleaning process was applied:

- **Dimensionality Reduction:**
  Irrelevant attributes, such as the data source, were removed to focus the model on predictive variables.
- **Unit Standardization & Price Normalization:**
  Mixed units (grams, pairs, pieces) were converted into base metrics (Kilograms, Liters, Single units). Prices were mathematically adjusted (e.g., multiplying 250g prices by 4) to ensure the model compares identical physical quantities.
- **Domain-Knowledge Filtering:**

  Beyond standard statistical outlier detection (IQR), we applied a Logical Filter at a 1,000 EGP threshold. This removed "Bulk/Wholesale" entry errors (e.g., Tea priced at 10,000 EGP) while preserving legitimate high-inflation price points (e.g., Meat at 500 EGP) essential for accurate forecasting.

Stage 3: Feature Engineering & Data Enrichment

To move beyond simple time-series analysis, we enriched the dataset with external economic "drivers":

- **Macroeconomic Indicators:** We integrated USD Exchange Rates and Fuel Prices, which are the primary catalysts for inflation in the Egyptian context.
- **Contextual Enrichment:** We engineered features for Religious Occasions (Ramadan and Eids) and Crisis Indicators (COVID-19 and the Russia-Ukraine War) using Hijri and Gregorian calendars.
- **Sequential Features (Lag Analysis):** We created "Lagged" variables and monthly inflation rates to capture the Autoregressive nature of food prices, where current prices are heavily influenced by the previous month's trends.

Stage 4: Data Transformation

- **Step: Categorical Encoding:** Applied Label Encoding for products and One-Hot Encoding for categories and crises.
- **Step: Cyclical Encoding (Sin/Cos):** Transformed months into cyclical waves. Standard numerical labels (1–12) incorrectly suggest that December and January are far apart. This cyclical approach mathematically links the end of the year (Month 12) back to the start (Month 1), ensuring the model recognizes their chronological proximity and preserves seasonal patterns.
- **Step: Smoothing**: Applied a 3-month moving average to the price.To remove "Noisy" short-term fluctuations and highlight the clear market trend.
- **Step: Z-Score Normalization:** Scaled numerical features to have a mean of 0 and a standard deviation of 1. This ensures that high-value features (like USD rate) do not mathematically overwhelm lower-value features during model training.

Stage 5: EDA and Statistical Insights (Analysis of Results)

Our Exploratory Data Analysis revealed critical insights into the Egyptian food market's mathematical behavior:

Distribution Shape (Skewness & Kurtosis):

Positive Skewness (2.46): The data shows a strong right-hand tail. This indicates that while most food prices remain at a baseline, there are frequent and significant "price surges" or spikes.

Leptokurtic Kurtosis (6.44): A kurtosis value much higher than 3 confirms a "Fat-tailed" distribution. This statistically proves that the Egyptian market is prone to extreme shocks and volatility rather than stable, incremental changes.

Correlation Matrix Analysis:

Price vs. Lagged Price (0.97): An extremely high correlation, confirming that the market has high momentum and that historical prices are the strongest predictors of future values.

Price vs. USD & Fuel (0.40 & 0.36): These coefficients show a moderate positive relationship, proving that currency devaluation and energy costs are significant secondary drivers of food inflation.

USD vs. Fuel (0.90): An exceptionally strong correlation (0.90) was observed between USD rates and Fuel prices, highlighting how energy costs in Egypt are almost directly tied to currency fluctuations

Descriptive Summary:

Post-standardization, the Mean (~0.00) and Standard Deviation (~1.00) confirm that our scaling was successful, providing a clean, zero-centered input for the machine learning models.

Data Modeling

This stage focuses on building a high-performance predictive system for food price forecasting. The methodology prioritizes chronological integrity and real-world economic interpretability.

1. Time-Aware Cross-Validation

To ensure the model is stable and not "lucky, we implemented a **Walk-Forward Time-Series Cross-Validation** (5 Folds).

- **Logic:** The model was tested on five different chronological "windows." This proves that the model's accuracy isn't just a result of the specific 2023 split, but is consistent across multiple time periods.
- **Metric Conversion:** Standardized error metrics (MAE) were transformed back into **Real EGP** values for every fold to provide a clear understanding of the model's performance in currency terms.

2. Methodology: Chronological 80/20 Split

Instead of a standard random split, a **Fixed Chronological Split** was implemented to prepare the final model.

- **Implementation:** The dataset was divided using a cut-off date of January 1st, 2023. This resulted in approximately **80% of the data (2018–2022)** being used for Training and **20% (2023–2026)** for Testing.
- **Rationale:** In time-series datasets, the order of events is critical. Shuffling data would allow the model to "see" future prices during training.
- **Impact:** This approach prevents **Data Leakage** by ensuring the model is trained only on historical data and evaluated on subsequent "future" data, mimicking real-world deployment scenarios.

3. Feature Selection & Leakage Prevention

A rigorous feature selection process was applied to maintain the scientific integrity of the predictions:

- **Target Variable:** Standardized Price.
- **Features:** USD Exchange Rates, Fuel Prices, Crisis Impact labels, Religious Occasions, and Product Categories.
- **Exclusions (Data Leakage Protection):** Columns such as Price_Smoothed and Monthly_Inflation_Rate were excluded from training. This prevents the model from "cheating" by using the answer key, forcing it instead to learn the actual economic drivers of price changes.

4. Algorithm Selection: XGBoost Regressor

- The **XGBoost (Extreme Gradient Boosting)** algorithm was selected for its advantages:
    - **Handling Non-Linear Economic Shifts:** Food prices in Egypt experience "shocks" and exponential jumps due to devaluation. XGBoost uses a forest of decision trees that can capture these non-linear relationships and complex interactions (e.g., the combined effect of Ramadan occurring during a period of high Fuel prices).
    - **Robustness to Diverse Feature Types:** Our dataset mixes numerical data (USD Rates) with categorical data (Product Categories). XGBoost handles these multi-

modal features efficiently and is naturally robust to the outliers common in volatile markets.
    - **Gradient Boosting Mechanism:** Unlike Random Forest, XGBoost uses an iterative process where each new tree corrects the errors of the previous ones. This allowed us to achieve high precision
- **Key Parameters:**
    - n_estimators=1000: Provides sufficient iterations for the model to learn deep patterns.
    - Early Stopping: Configured to stop training automatically if the "Test Loss" stops improving, preventing the model from simply memorizing the training data.
    - Regularization (Lambda/Alpha): Applied to keep the model "simple" and prevent overfitting to extreme price spikes.

5. Performance Evaluation & Diagnostics

The model is evaluated using three primary statistical metrics across both Training and Unseen Testing data:

- R^2 **(Coefficient of Determination):** Measures the percentage of price variance explained by the model.
- **MAE (Mean Absolute Error):** Calculated in both standardized units and real currency to measure average prediction deviation.
- **RMSE (Root Mean Squared Error):** Used to penalize larger errors more heavily, ensuring the model remains reliable during economic shocks.
- **Loss Curve Analysis:** A visual diagnostic of RMSE progression was utilized to identify the "Overfitting Gap," ensuring the model generalizes well to future market conditions.

6. Explainability & Feature Importance

A key component of professional data science is **Interpretability**. The pipeline extracts the **Feature Importance** scores to identify which variables (e.g., USD rates vs. previous month's price) are the primary drivers of food inflation in the dataset. This adds a layer of transparency to the automated predictions.

6. Robust Integrity Audit

To conclude the modeling stage, a **Product Distribution Audit** was performed to verify the dataset split.

- **Audit Goal:** To ensure that every one of the 61 products appeared in both the training set (pre-2023) and the testing set (post-2023).

- **Impact:** This confirms that the model has "learned" the historical behavior of 100% of the products before being asked to forecast their future prices, eliminating "blind" predictions.

Visualization & Interpretation

A: Pre-Training Exploratory Data Analysis (EDA)

Focuses on understanding data quality, distributions, and baseline patterns before modeling.

1. **Price Trend Analysis (Top Products):** Tracks historical fluctuations of the most frequent items to identify long-term volatility and general market direction.
2. **Product Category Distribution:** Analyzes the volume of data per category (e.g., Meat, Dairy) to ensure dataset balance and identify dominant market sectors.
3. **High-Value Product Ranking:** Highlights the top 10 most expensive items by historical average, identifying commodities that consume the largest portion of consumer budgets.
4. **Seasonal Price Pattern Analysis:** Utilizes monthly boxplots to detect recurring volatility cycles linked to specific times of the year (e.g., agricultural seasons or religious occasions).
5. **Data Smoothing Validation (Actual vs. Smoothed):** A critical visual audit to confirm that noise reduction techniques effectively cleaned the data while preserving the core price signals.
6. **Anomaly Detection:** Pinpoints extreme outliers and price spikes caused by data entry errors or unprecedented market shocks that require specific preprocessing.
7. **Correlation Analysis Matrix:** A heatmap measuring the statistical strength between variables (Fuel, USD, Lags) to identify the primary macro-economic drivers of price change.

Group B: Model Evaluation & Performance Metrics

Focuses on assessing the model's learning process and its ability to generalize to unseen data.

1. **Model Training Progress (Loss Curve):** Monitors the RMSE (Root Mean Squared Error) across iterations to ensure the model is converging correctly and to detect potential Overfitting.
2. **Prediction Accuracy (Actual vs. Predicted):** Directly compares the "Ground Truth" prices against the model's forecasts on the test set to evaluate real-world predictive precision.

Group C: Post-Training & Insight Analytics

Focuses on interpreting the "Why" behind price movements and the impact of external economic drivers.

1. **Economic Crisis Impact Assessment:** Evaluates how major shocks (e.g., COVID-19, Ukraine War) shifted price trajectories, identifying which events acted as the strongest inflationary catalysts.
2. **Exchange Rate Impact Analysis:** Examines the direct sensitivity of local food prices to USD fluctuations, quantifying the market's dependence on foreign currency stability.